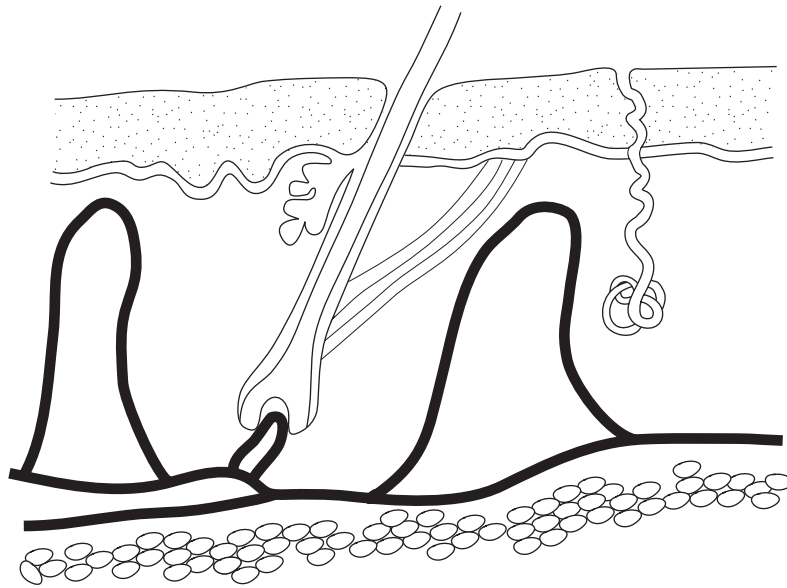


6. The diagram shows a section through human skin.



(a) On the diagram, use arrows to label the:

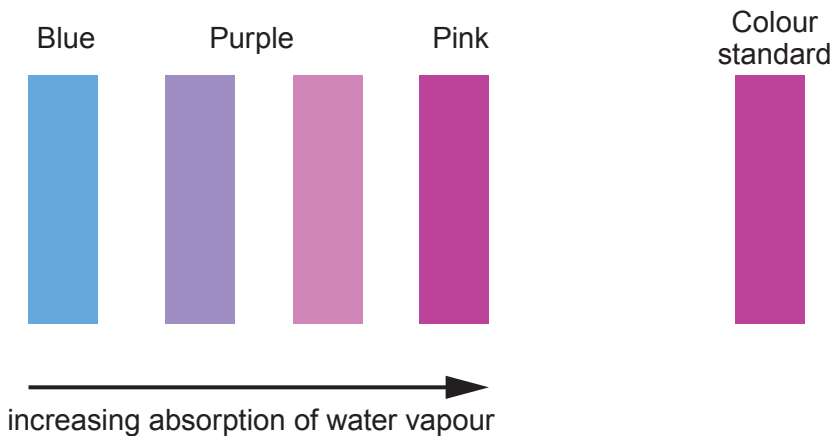
[2]

- I. sweat gland;
- II. erector muscle.



(b) Some GCSE students decided to investigate the effect of temperature on sweating.
This is their method.

Volunteers sat in a small room, the temperature of which could be finely controlled.
Strips of cobalt chloride paper were placed in small wire baskets which were held 2 mm above the surface of the skin of the arm of the volunteers.
Cobalt chloride paper turns from a blue to purple to pink colour in the presence of water vapour, as shown below.



The time taken for the cobalt chloride paper to change to the colour of the standard was recorded. The results are shown in the table below.

		Time taken for cobalt chloride paper to change to colour of standard (s)			
Volunteer No.	Room temp (°C)	Trial No. 1	Trial No. 2	Trial No. 3	Mean
1	15	363	387	376	375
2	20	321	304	309	311
3	25	287	289	265	280
4	30	221	230	202	218
5	35	187	176	167	

- (i) **Complete the table** by calculating the mean time for volunteer number 5. [2]
- (ii) Using these results only, what conclusion can be drawn about the effect of temperature on sweating? [1]

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Examiner
only

- (iii) Alun, one of the boys in the class, said that *‘we can also conclude, from the results, that the effect of temperature differs in different people’*.
The rest of the class disagreed with this. State why they disagreed. [1]

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- (iv) Explain why the final colour change of the cobalt chloride paper is compared with a standard. [1]

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- (c) Explain the source of the water vapour which turned the cobalt chloride paper pink. [2]

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- (d) Melanoma is a type of skin cancer. Briefly describe how cancer occurs in the body. [1]

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